

Threat Hunting Process Lab Report

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Course: Network Forensics and Threat Hunting

Lab: Threat Hunting Process (PCAP: 2021-02-08-traffic-analysis-exercise.pcap)

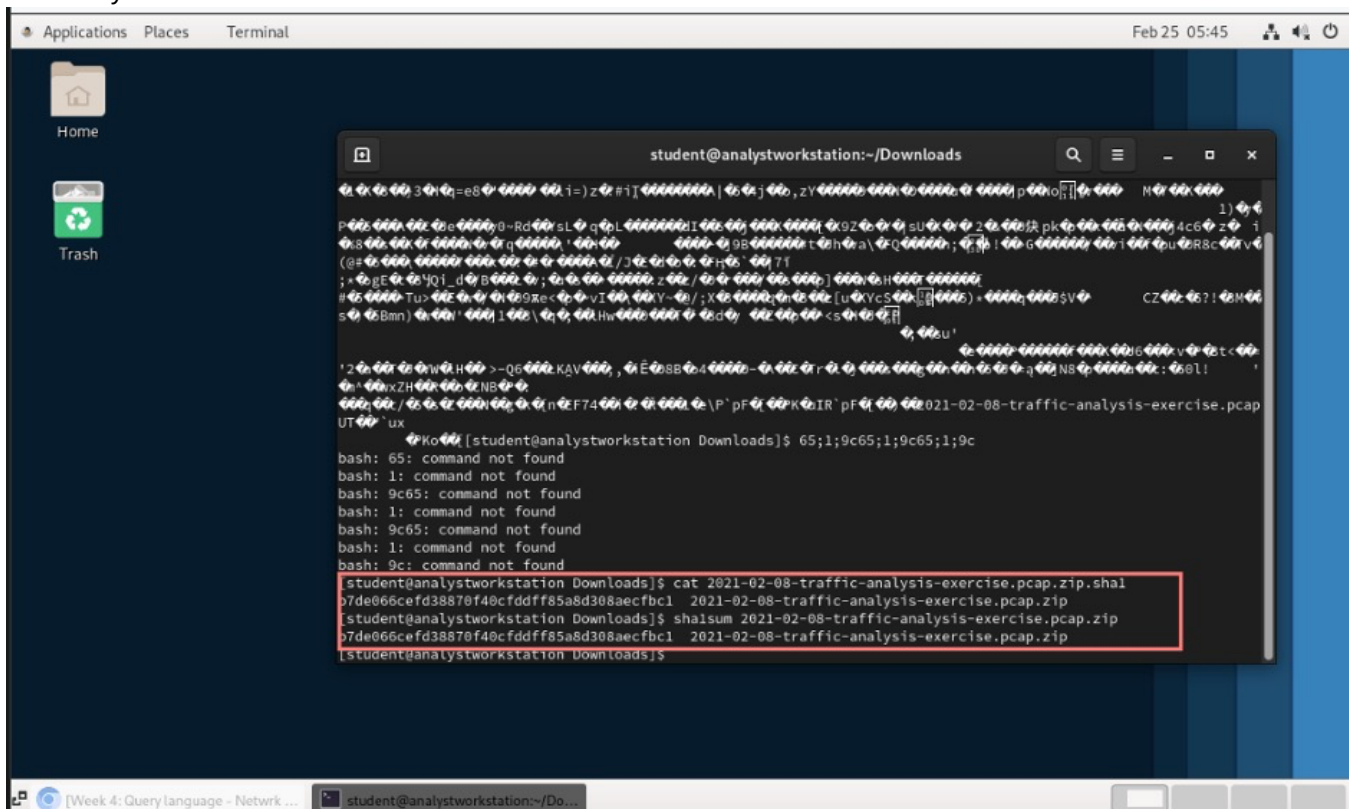
Date: Feb 25, 2026

Objective

Verify the integrity of the provided PCAP, import it into Security Onion, and perform an alert-driven threat hunt. The goal is to document key alerts, gather evidence, and summarize the most likely attack activity.

Step 0 — Verify & Import PCAP

I validated the PCAP ZIP file using the provided SHA1 hash and then imported the PCAP into Security Onion for analysis.



The screenshot shows a terminal window titled 'student@analystworkstation:~/Downloads'. The terminal output displays the SHA1 hash of the file '2021-02-08-traffic-analysis-exercise.pcap.zip' and compares it with the provided hash. The output is as follows:

```
student@analystworkstation Downloads$ cat 2021-02-08-traffic-analysis-exercise.pcap.zip.sha1
57de666cfd38870f40cfdff85a8d308aecfbc1 2021-02-08-traffic-analysis-exercise.pcap.zip
student@analystworkstation Downloads$ sha1sum 2021-02-08-traffic-analysis-exercise.pcap.zip
57de666cfd38870f40cfdff85a8d308aecfbc1 2021-02-08-traffic-analysis-exercise.pcap.zip
student@analystworkstation Downloads$
```

The output shows that the SHA1 hash of the file matches the provided hash, indicating that the file is authentic and has not been tampered with.

Figure 1. SHA1 verification (hash matches).

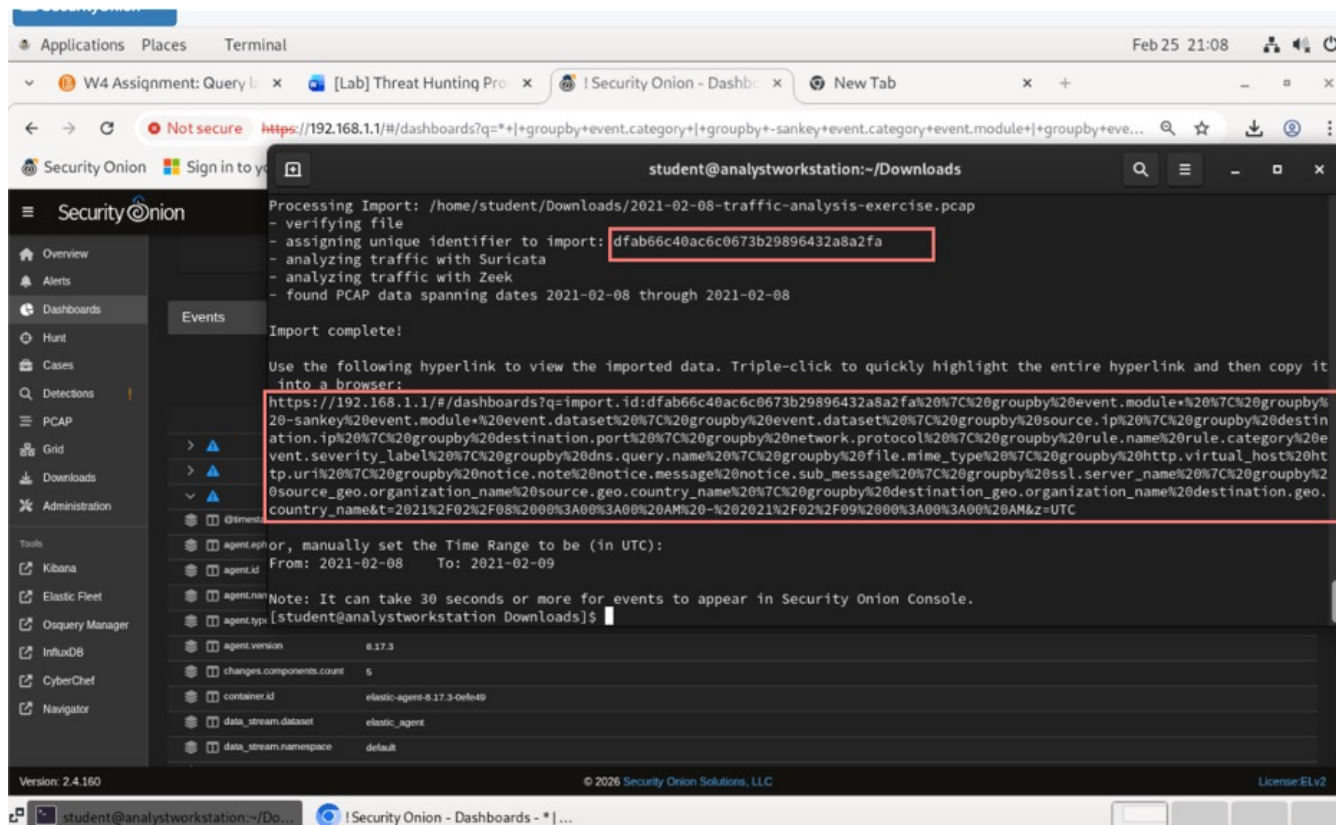


Figure 2. Security Onion PCAP import completed (import id and dashboard link).

Step 1 — Know Your Network (Baseline)

Item	Observed/Assumed Value
LAN range	10.2.8.1 – 10.2.8.254
CIDR / Mask	10.2.8.0/24
Gateway	10.2.8.1
Broadcast	10.2.8.255
Domain	ascolimited.com
Domain Controller / Local DNS	10.2.8.2
Primary suspected workstation	10.2.8.101

Step 2 — Alert-Driven Threat Hunt (4 Alerts)

I reviewed Suricata alerts and pivoted into Hunt to capture supporting evidence for each alert. For each alert, I documented a hypothesis, key evidence, and a conclusion.

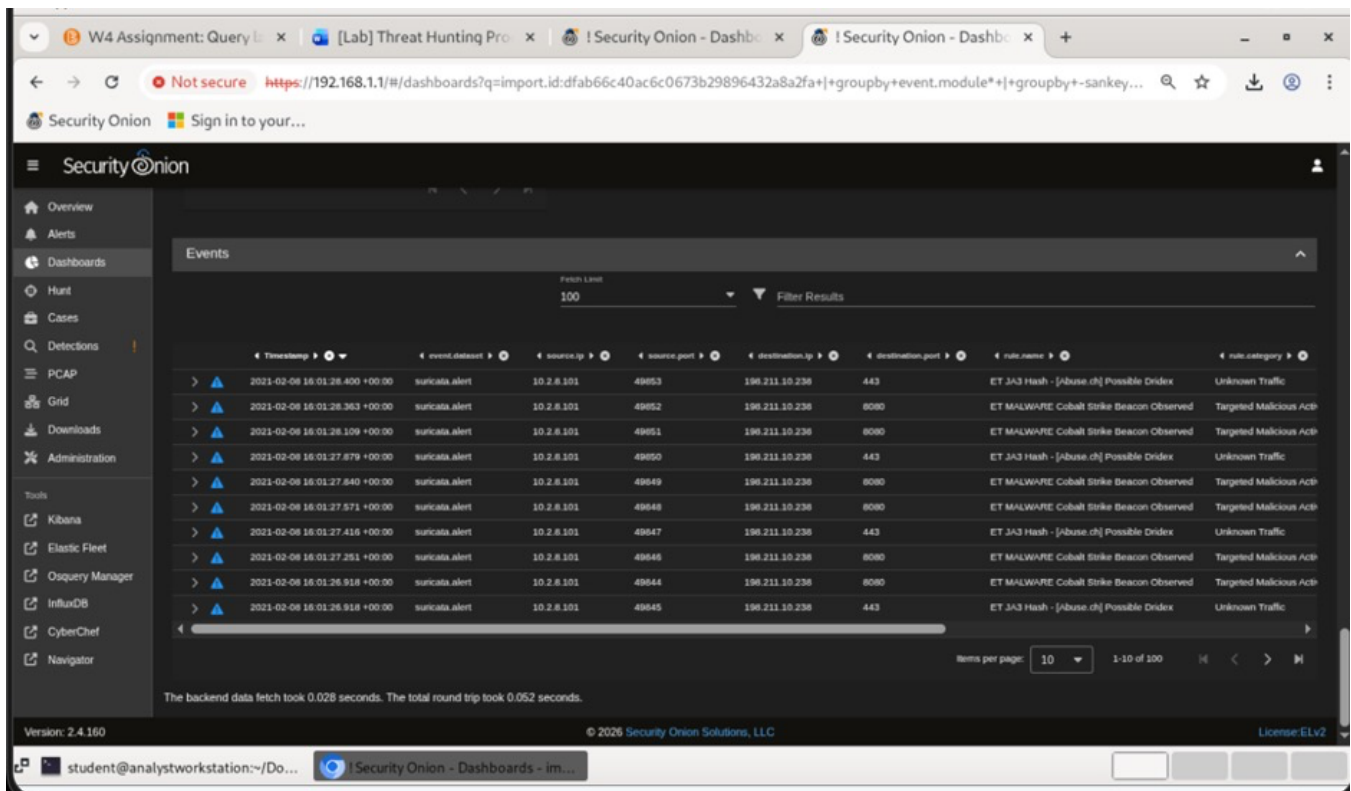


Figure 3. Events view showing multiple related Suricata alerts.

Alert 1 — ET MALWARE Cobalt Strike Beacon Observed

Hypothesis: Host 10.2.8.101 is compromised and beaconing to an external command-and-control (C2) server.

Evidence: The alert details show outbound traffic from **10.2.8.101** to **198.211.10.238** on **port 8080** with the signature “Cobalt Strike Beacon Observed.”

Conclusion: Confirmed. The specific IDS signature and consistent outbound connection pattern strongly indicate active C2 beaconing.

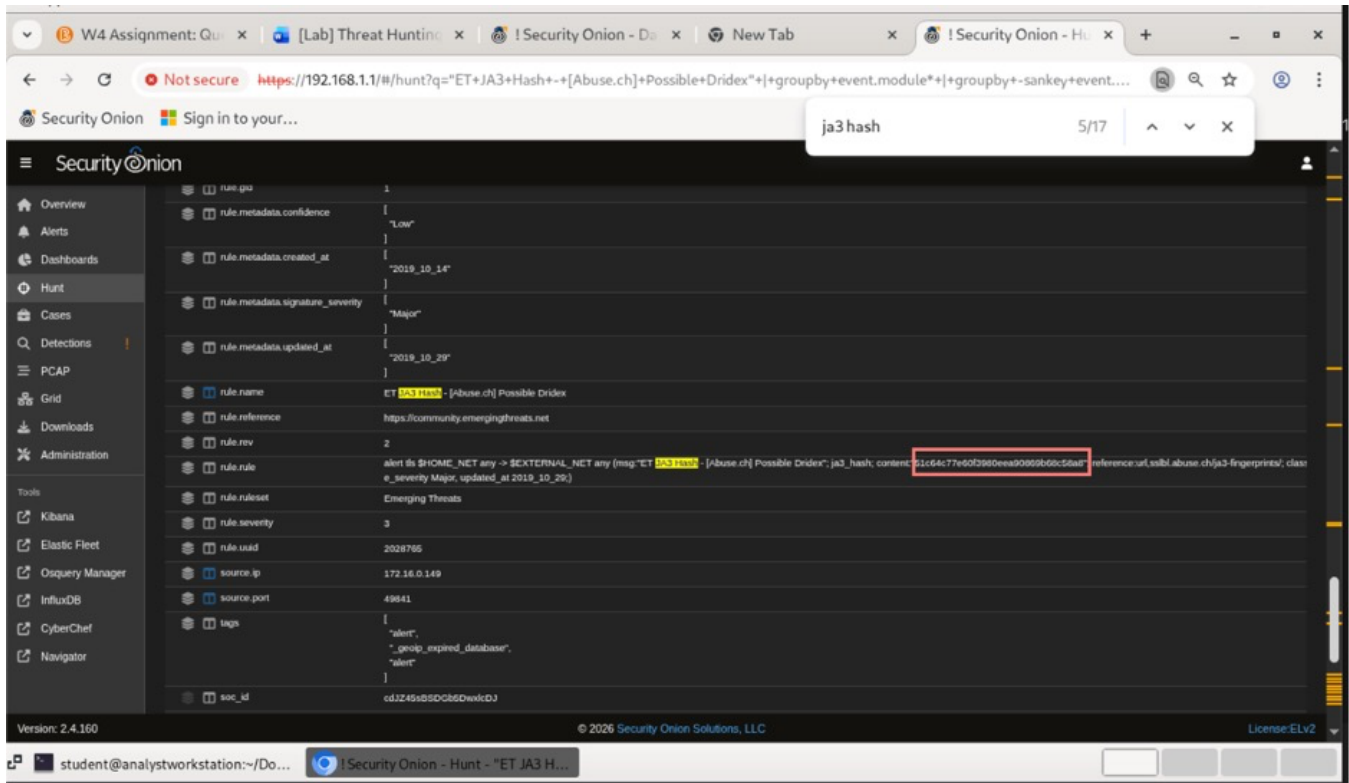


Figure 5. Hunt view of repeated JA3/Dridex-related connections.

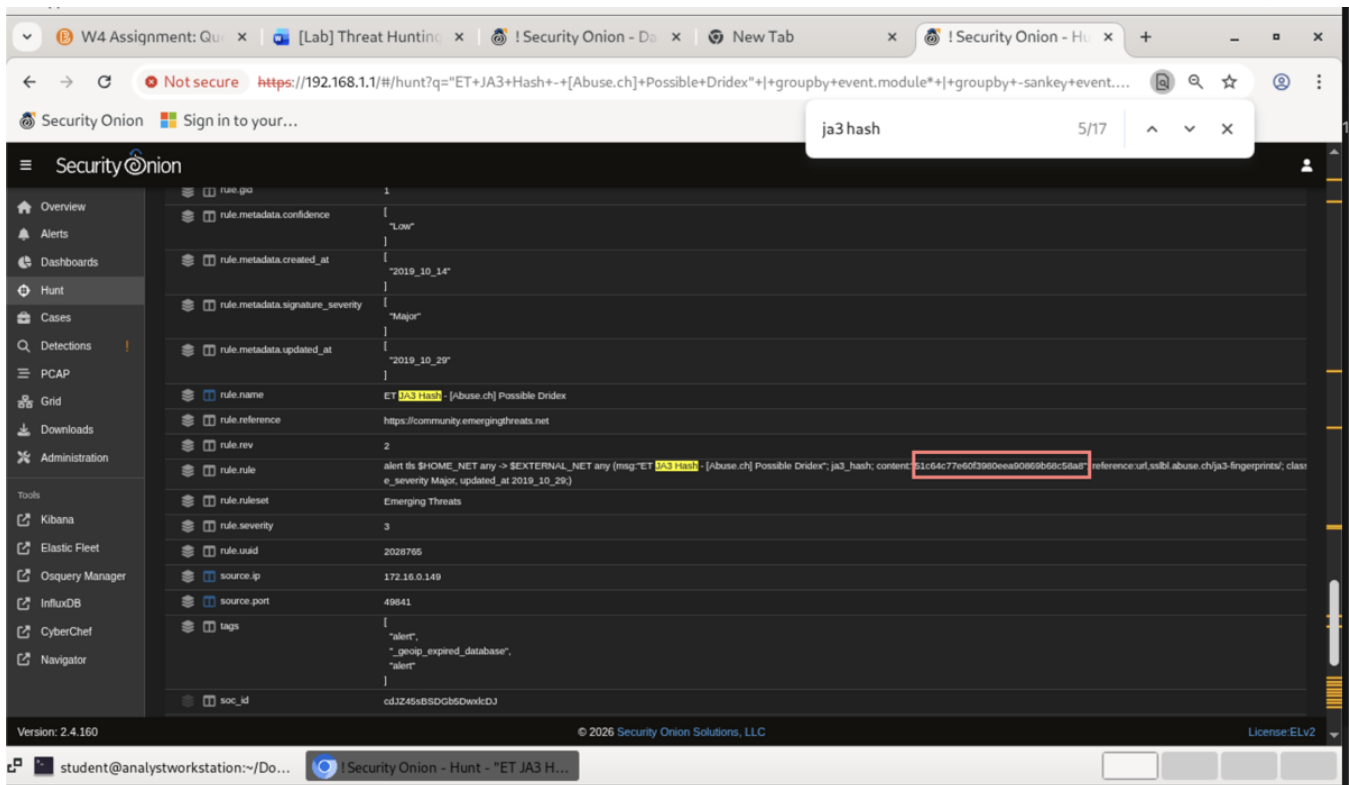


Figure 6. Alert details showing the JA3 hash value used by the rule.

Alert 3 — ET PHISHING Let's Encrypt SSL Cert with IDN/Punycode Domain

Hypothesis: A phishing domain is using a punycode (IDN) lookalike and a Let's Encrypt certificate to appear legitimate.

Evidence: The event contains the punycode domain **xn--vigatorsky-key-if2g.com** and is flagged as "Possible Phishing" with Let's Encrypt certificate context.

Conclusion: Confirmed suspicious. Punycode lookalike domains with free SSL certificates are common in phishing infrastructure.

Figure 7. Phishing alert details showing punycode domain (xn--...).

Alert 4 — ET INFO External IP Lookup (ipify.org)

Hypothesis: The host is attempting to discover its public IP address (recon step often seen in malware workflows).

Evidence: The alert indicates a DNS lookup to **ipify.org** and shows destination IP **54.235.147.252**, consistent with an external IP-check service lookup.

Conclusion: Confirmed. This behavior supports reconnaissance / environment awareness prior to or during external communications.

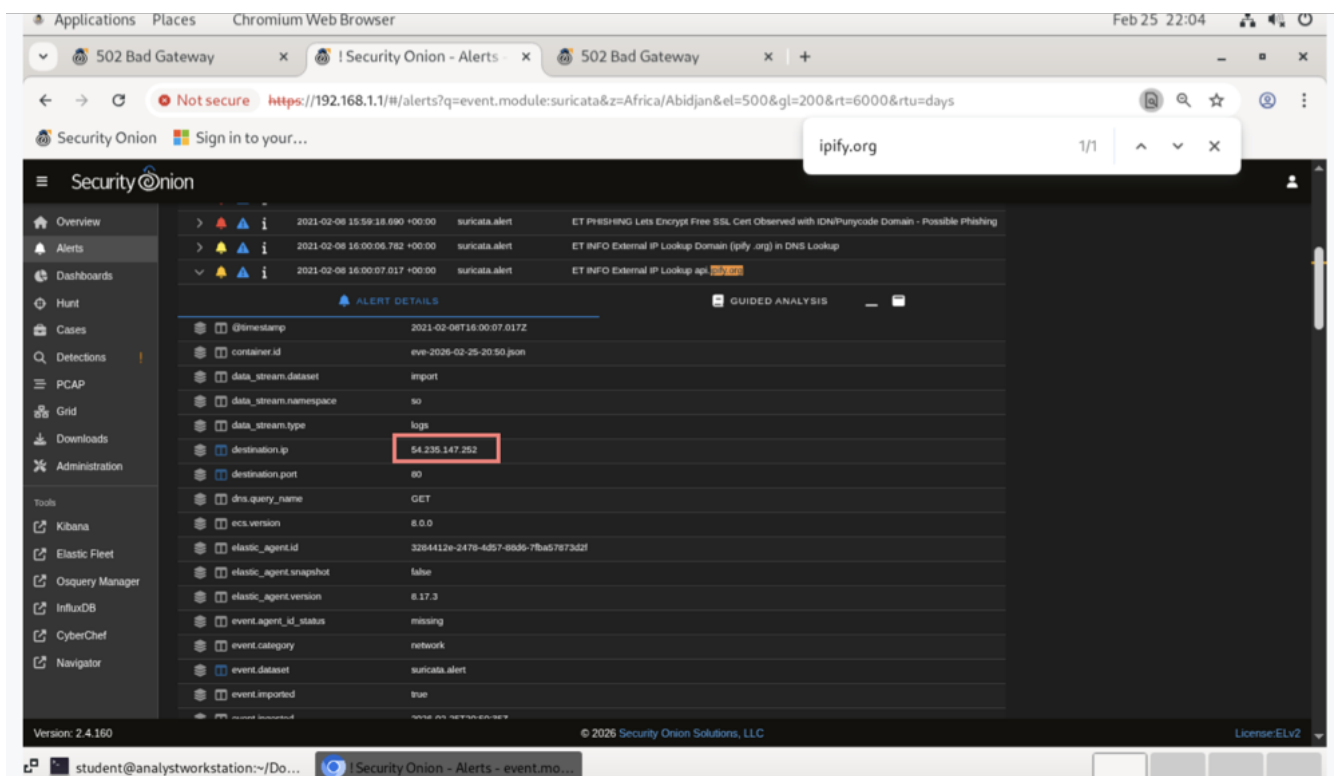


Figure 8. ipify.org external IP lookup alert details.

Overall Assessment

Across the four alerts, the evidence points to likely compromise of workstation **10.2.8.101**. The strongest indicator is the confirmed **Cobalt Strike** beaconing to **198.211.10.238:8080**. Additional supporting indicators include a **Dridex-associated JA3** fingerprint and interaction with a **punycod phishing domain**, along with an **external IP lookup** via ipify.org. Taken together, these artifacts strongly suggest malicious activity and command-and-control communications.